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From OLTP to OLAP: Is PostgreSQL Eating Analytics Too?

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Background



- Co-Founder of benchANT
- PhD about benchmarking cloud and database systems
- All about distributed systems, databases and cloud
- Enjoys to demystify the black art of database benchmarking
- Loves every kind of racket sports

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How Analytics Vendors Building on PostgreSQL?

Databricks Agrees to Acquire Neon to Deliver Serverless Postgres for Developers + AI Agents

May 14, 2025



Mooncake Labs joins Databricks to accelerate the vision of Lakebase



Today we're excited to announce our partnership with ClickHouse: Managed Postgres as part of the ClickHouse Cloud platform, powered by Ubicloud Postgres. ClickHouse customers can now use Postgres for their transactional workloads, tightly integrated with ClickHouse for analytics.

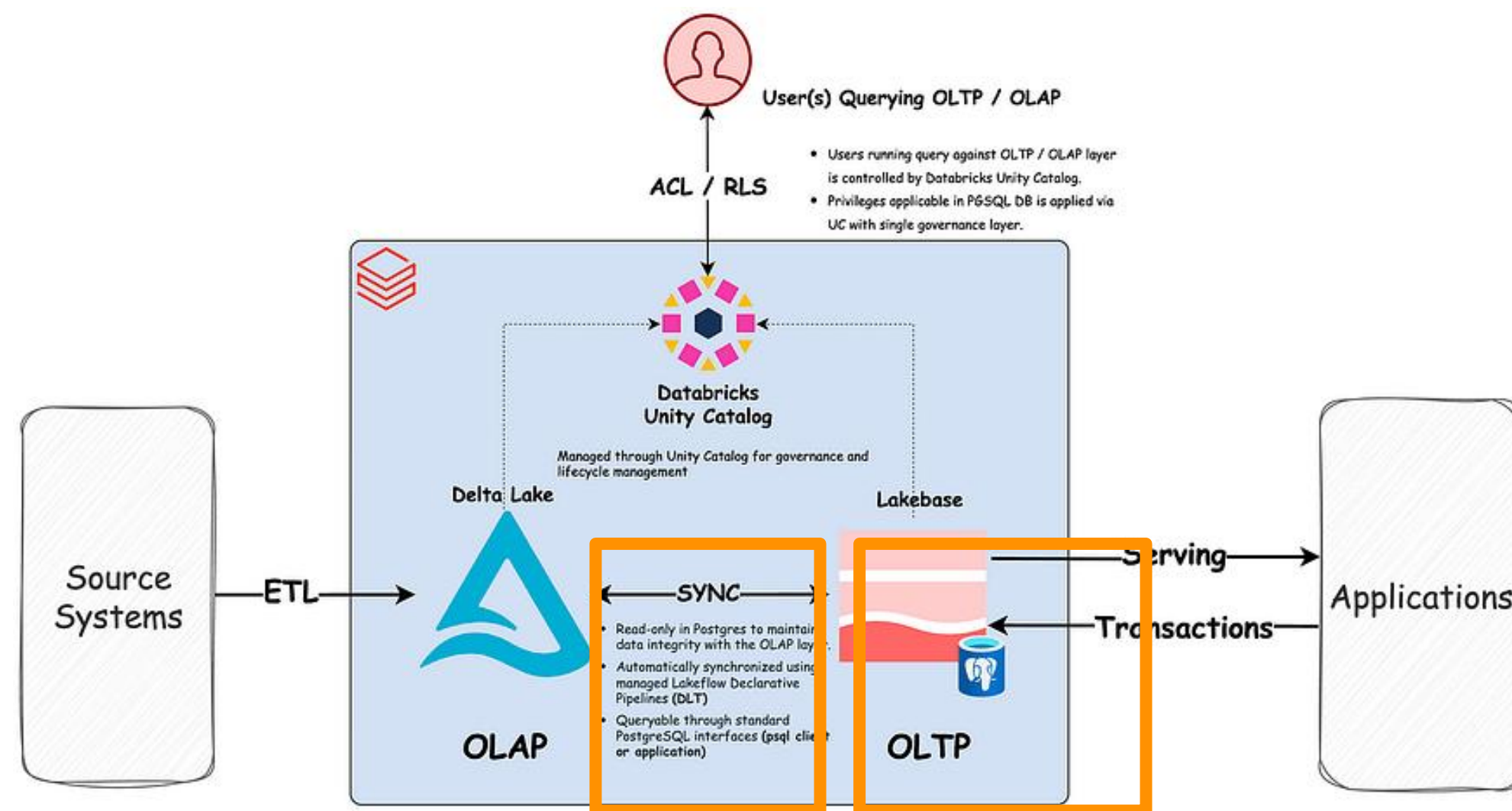
This gives you a unified data stack: Postgres for transactions, ClickHouse for analytics, with native capabilities to sync and query data across datastores. Ubicloud Postgres, backed by fast, local NVMe storage, powers the Postgres experience.

<cloud> ubicloud × |||| ClickHouse

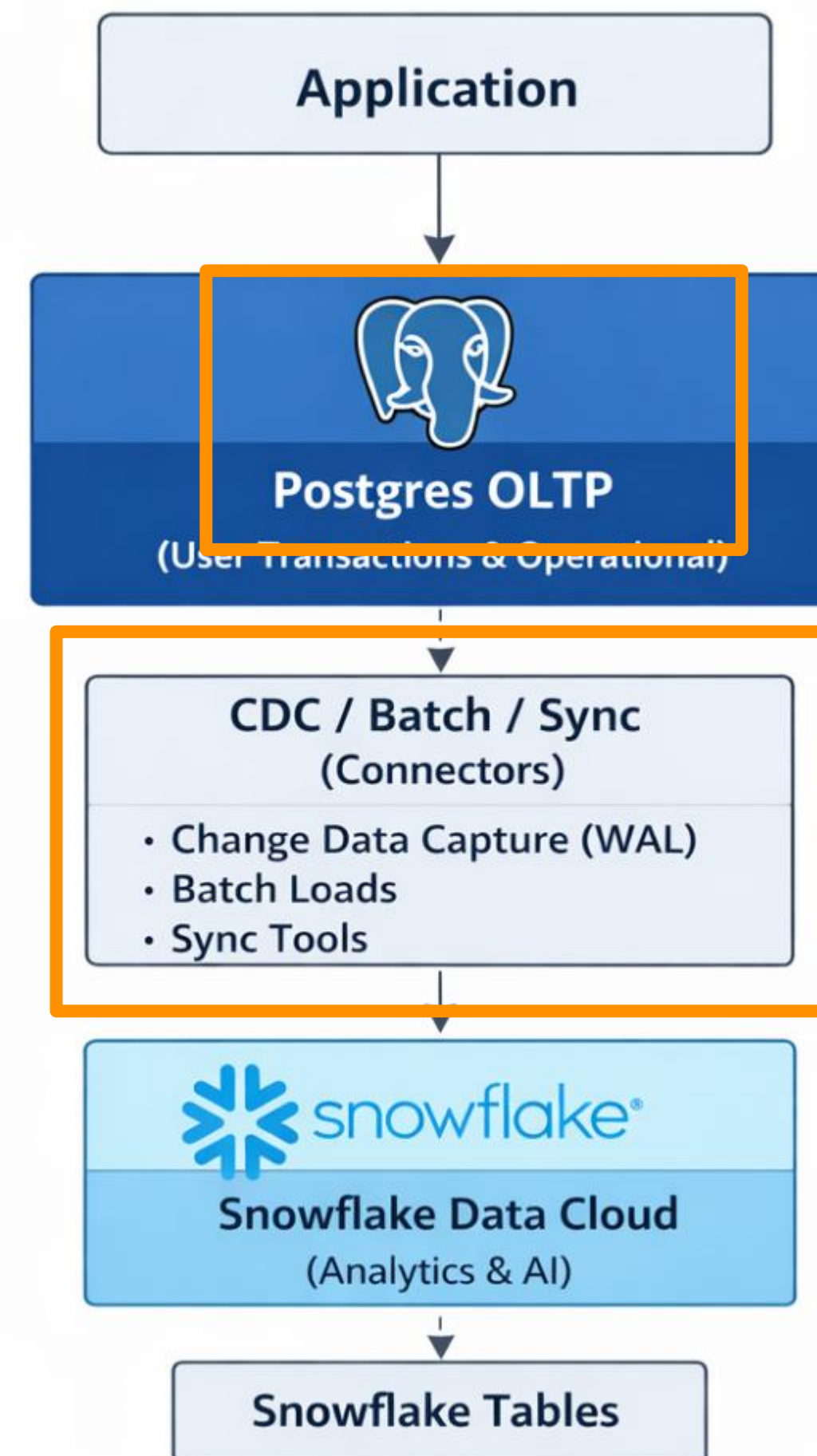


Snowflake Acquires Crunchy Data to Bring Enterprise Ready Postgres Offering to the AI Data Cloud

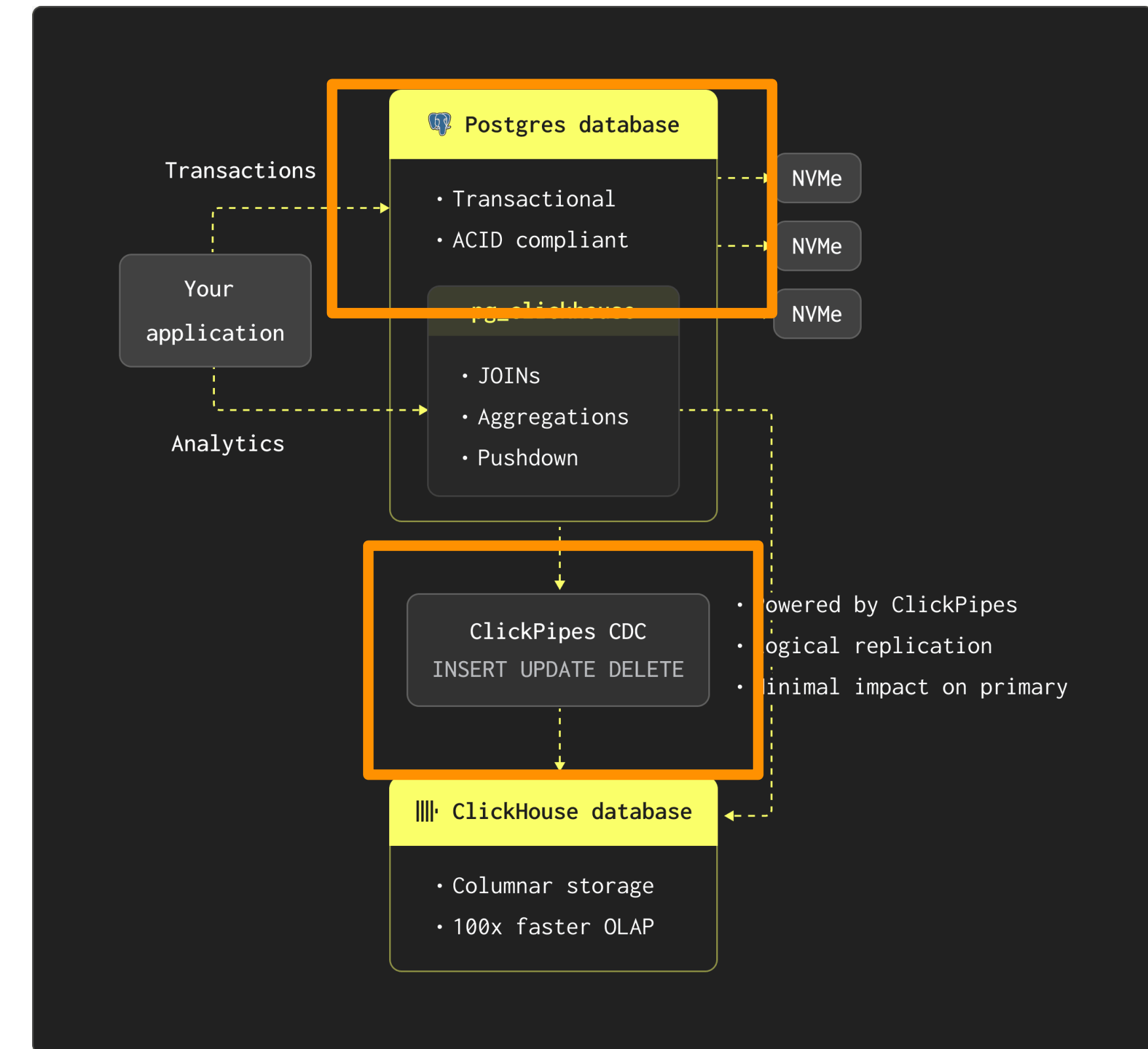
How Analytics Vendors Building on PostgreSQL?



<https://medium.com/@nivethanvenkat28/ready-to-give-your-database-ai-wings-meet-databricks-lakebase-your-new-pilot-part-1-4067590a4f71>

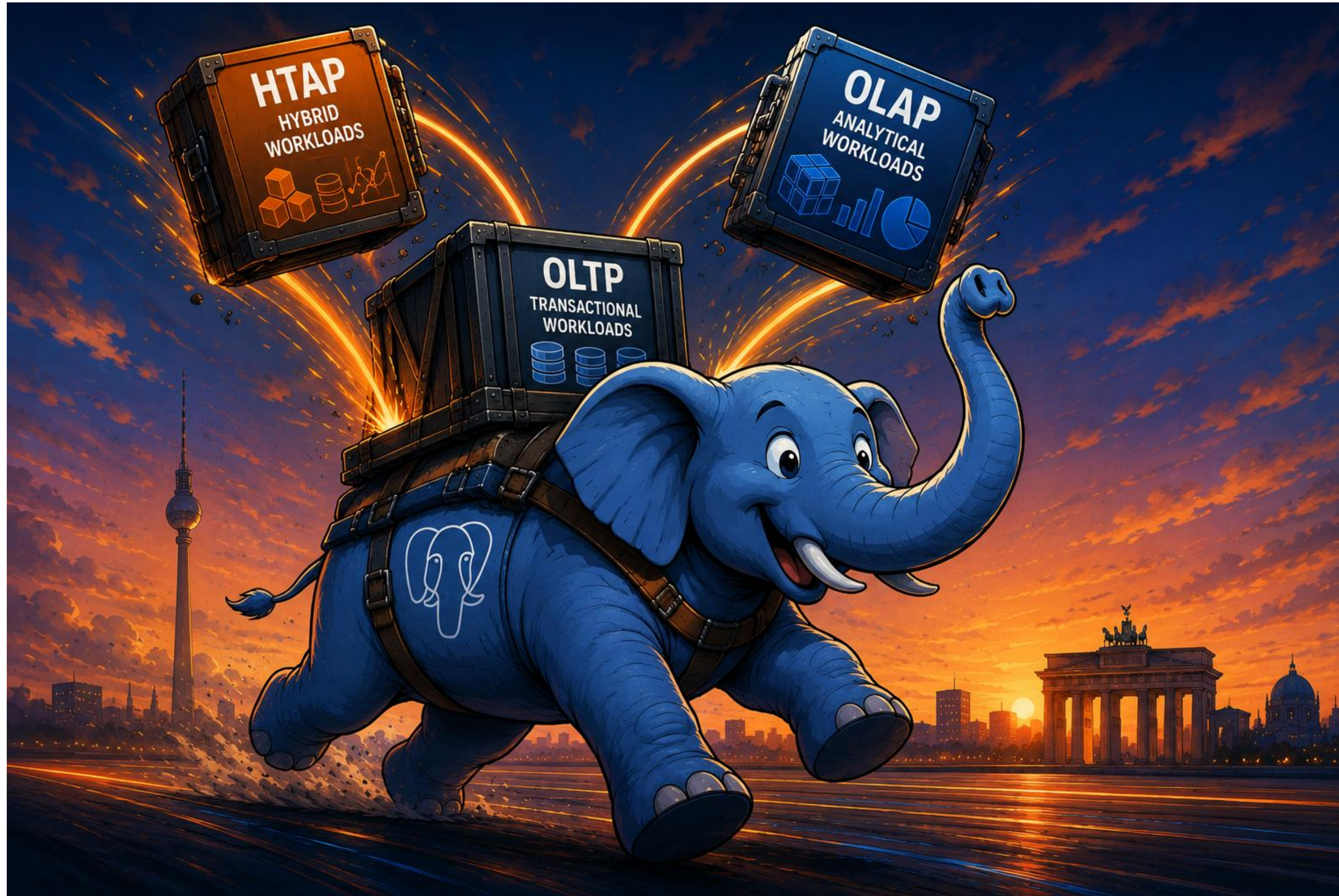


<https://www.linkedin.com/pulse/postgres-snowflake-better-together-snowflake-computing-wwlwc/>



<https://clickhouse.com/blog/postgres-managed-by-clickhouse>

Why Are Analytics Vendors Betting on PostgreSQL?



What Makes an Analytical Workload?

- Analytical domains: data warehouse, business intelligence, ad-hoc analytics, dashboard analytics, real-time analytics, petabyte analytics, search, agentic (?) ...
- Data set size: disk bound, 100 GB → PB
- Data schema: normalized vs. denormalized
- Query types: complex including joins, aggregations, grouping
- Data access: read-dominated with batched/streamed inserts, depending on the application domain (e.g. real-time analytics)

Can we turn PostgreSQL into an Analytical Database?

Distributed Columnar Storage with Citus

- Actively maintained PostgreSQL extension with > 12.5k GitHub stars (Citus Data acquired by Microsoft in 2019)
- Citus primary focus is to turn PostgreSQL into a distributed database by adding distributed tables and a distributed query engine
- It also adds support for a columnar storage engine that compresses data, speeds up scans, and supports fast projections, both on regular and distributed
- Citus columnar is the successor of cstore_fdw
- More details on <https://github.com/citusdata/citus/tree/main> and <https://github.com/citusdata/citus/tree/main/src/backend/columnar>

Real-Time Analytics with TimescaleDB

- Actively maintained PostgreSQL extension with > 22.8k GitHub stars, maintained by Tiger Data Inc.
- TimescaleDB primary focus is to enable high-performance time-series workload handling in PostgreSQL
- TimescaleDB also adds support for a columnar storage engine as time-series workloads can benefit from columnar storage, especially with high compression and faster vectorized queries
- More details on <https://github.com/timescale/timescaledb>

Embedding **DuckDB** into PostgreSQL

- Hydra started by building a columnar engine for PostgreSQL from scratch, now maintaining the pg_duckdb extension (3.1k GitHub stars) in collaboration with MotherDuck Inc.
- DuckDB is a lightweight embedded database that provides a columnar and vectorized execution engine for analytical workloads
- pg_duckdb integrates DuckDB's columnar-vectorized analytics engine into PostgreSQL
 - Improves performance for aggregations, window functions, complex JOINS
 - **Execute analytics queries without changes to query or data export**
- More details on https://github.com/duckdb/pg_duckdb

Why We Chose **pg_duckdb**

- Current release is v1.1.1 (released Dec 18, 2025)
- Execute analytics queries without changes

```
SET duckdb.force_execution = true;  
SELECT  
    order_date,  
    COUNT(*) AS number_of_orders,  
    SUM(amount) AS total_revenue  
FROM  
    orders  
GROUP BY  
    order_date  
ORDER BY  
    order_date;
```

- Read/write data from/to data lakes
- Integration with cloud analytics based on MotherDuck

Benchmarking PostgreSQL Analytics

Benchmark Objectives

1. Is pg_duckdb the silver bullet to push PostgreSQL analytics performance to the ClickHouse level?
2. How does PostgreSQL with pg_duckdb compete against ClickHouse for analytical workloads of the data warehouse/BI and the dashboard/ad-hoc analytics domain?
3. What is the performance improvement of pg_duckdb over native native PostgreSQL?

Technical Benchmark **Setup**

- PostgreSQL Version 17 with an analytical workload config by <https://pgtune.leopard.in.ua/>
 - pg_duckdb v1.1.1
 - AWS EC2 m7i.2xlarge VM with 8 vCores/32GB RAM and 500GB GP3 storage
- Clickhouse Cloud version 25.12 on AWS with 2 nodes of 4 vCores/16 GB RAM
- Benchmark instance is deployed on a c6i.4xlarge VM in the same region the PostgreSQL instance
- Full data set on GitHub: <https://github.com/benchANT/berlin-buzzwords-from-OLTP-to-OLAP-is-PostgreSQL-eating-analytics-too>

ClickBench Setup

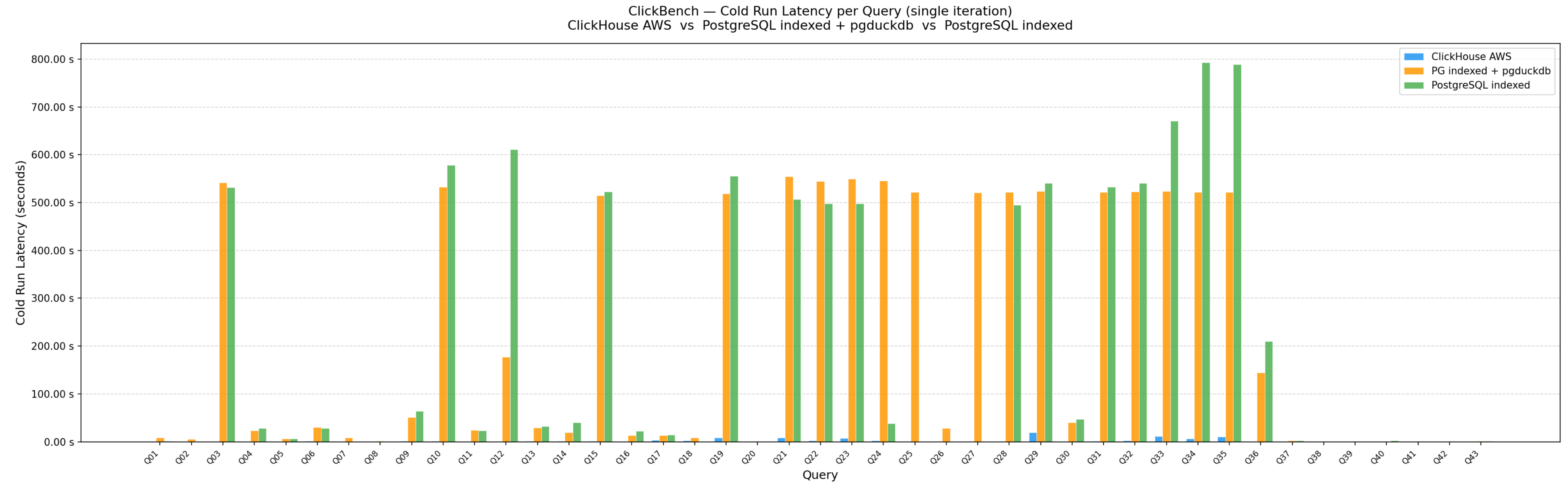
- ClickBench data set from <https://github.com/ClickHouse/ClickBench/tree/main>
- PostgreSQL schema, indexes and queries from <https://github.com/ClickHouse/ClickBench/tree/main/postgresql-indexed>
- PostgreSQL + pg_duckdb schema, indexes and queries from https://github.com/ClickHouse/ClickBench/blob/main/pg_duckdb-indexed/queries.sql
- Benchmark execution and metric reporting is integrated into the benchANT platform but follows the ClickBench methodology

ClickBench **Data Set & Methodology**

- Analytics domain: dashboard & BI
- Data set size is ~ 100M rows
- Flat table with 104 columns
- 43 analytical queries with static values focusing on scan, filter, aggregation, grouping, expressions, string processing
- Single threaded and sequential query execution
- Hot vs. warm vs. cold runs

ClickBench — Is pg_duckdb the silver bullet to push Post analytics performance to the ClickHouse level?

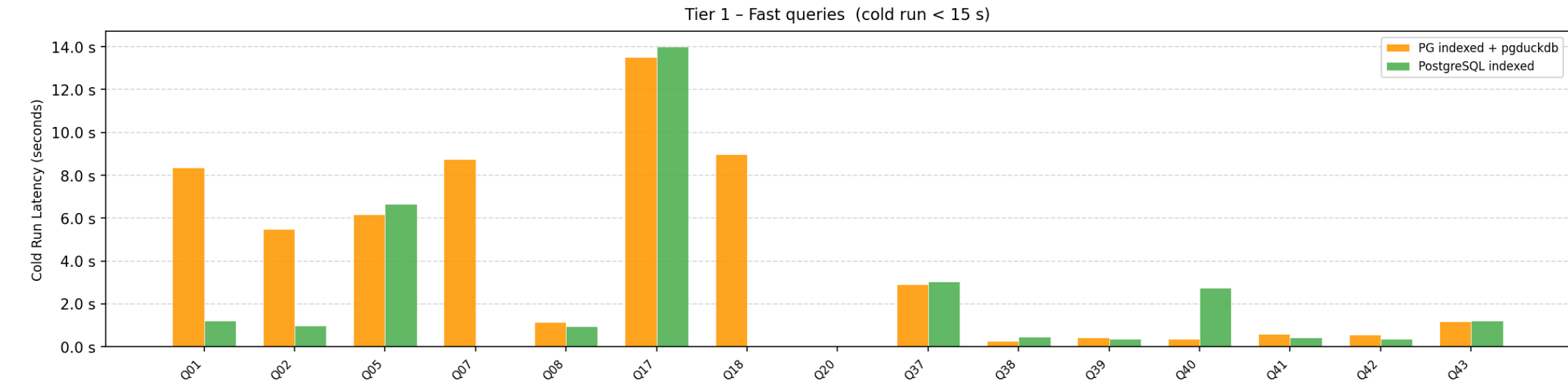
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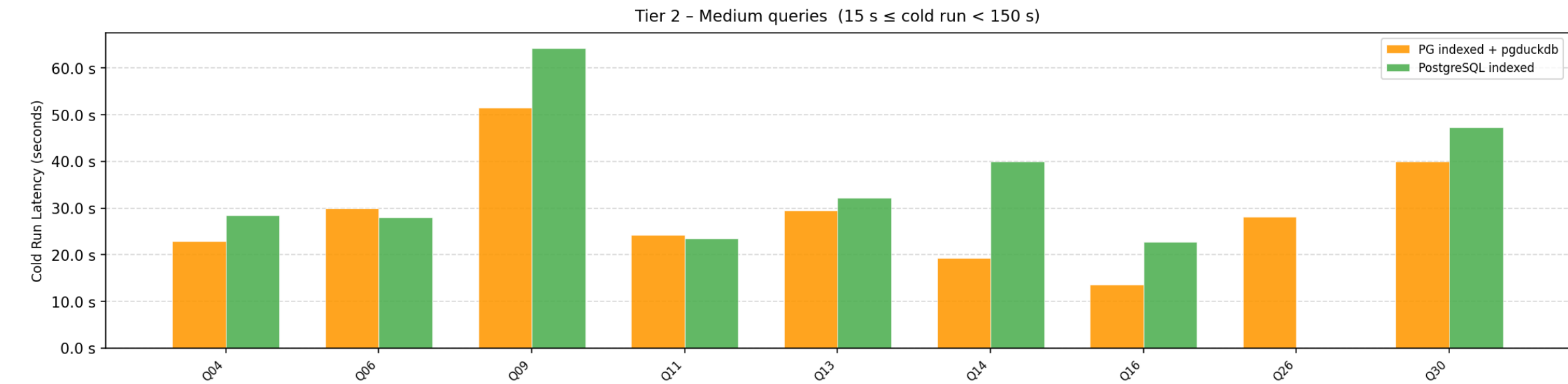
No! Yet, expecting one PostgreSQL extension to turn PostgreSQL into ClickHouse overnight may have been a little optimistic.

ClickBench — What is the performance improvement of pg_duckdb over native native PostareSQL

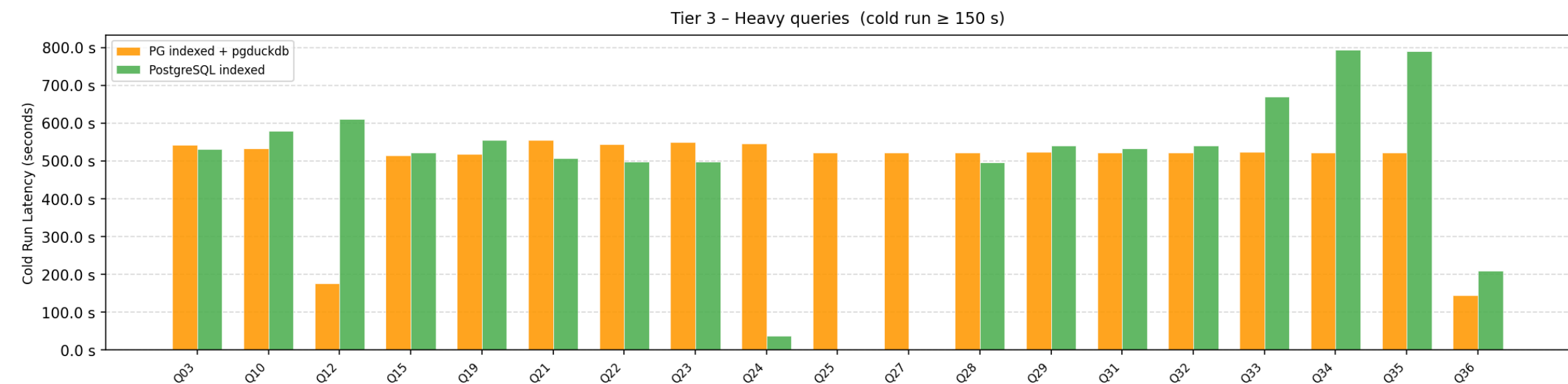
ClickBench — Cold Run Latency per Query (single iteration, indexed)
PostgreSQL + pgduckdb vs PostgreSQL indexed



	pg_duckdb	PG
✓ Wins	2	9
Draws	4	



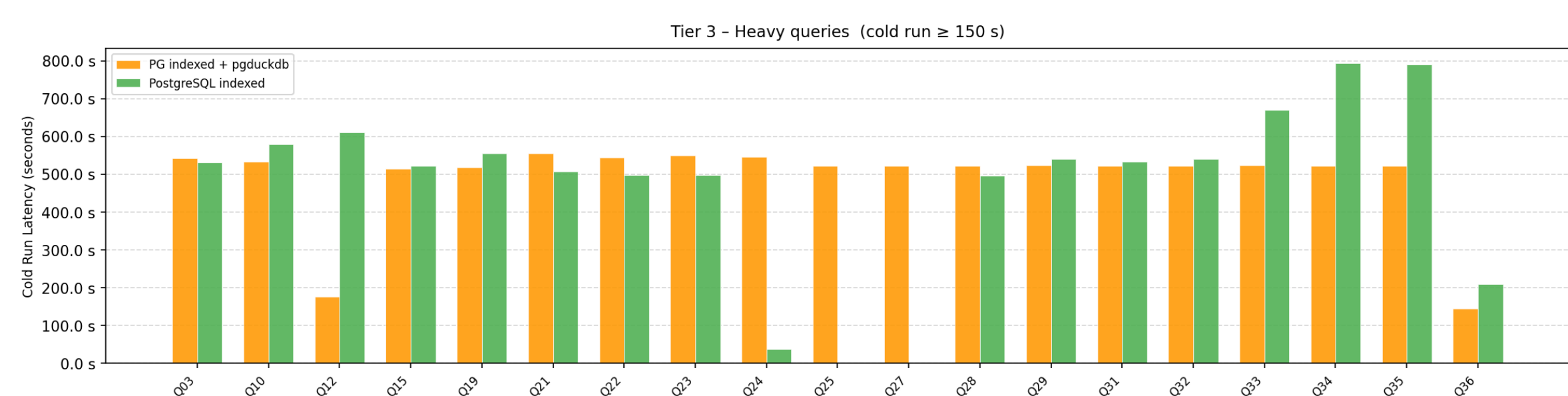
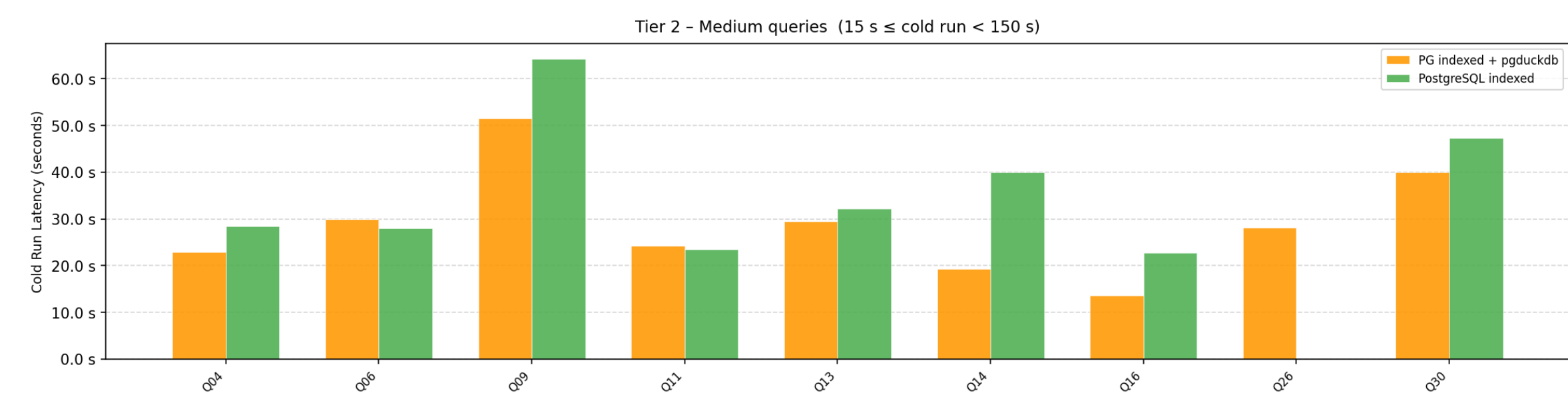
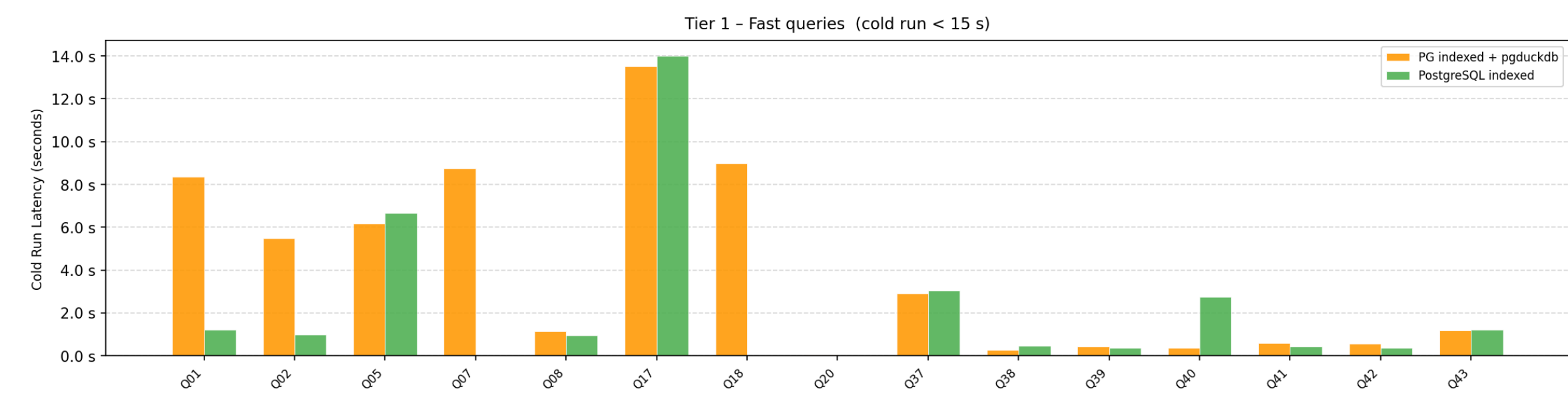
	pg_duckdb	PG
✓ Wins	5	1
Draws	3	



	pg_duckdb	PG
✓ Wins	5	3
Draws	11	

ClickBench — What is the performance improvement of pg_duckdb over native native PostgreSQL

ClickBench — Cold Run Latency per Query (single iteration, indexed)
PostgreSQL + pgduckdb vs PostgreSQL indexed



	Query	pg_duckdb	PG	Factor
✓ Best	Q40	0.36s	2.75s	7.6x faster
✓ Best	Q38	0.27s	0.47s	1.73x faster
✗ Worst	Q18	8.98s	0.0009s	~9,900x slower
✗ Worst	Q07	8.77s	0.008s	~1,163x slower

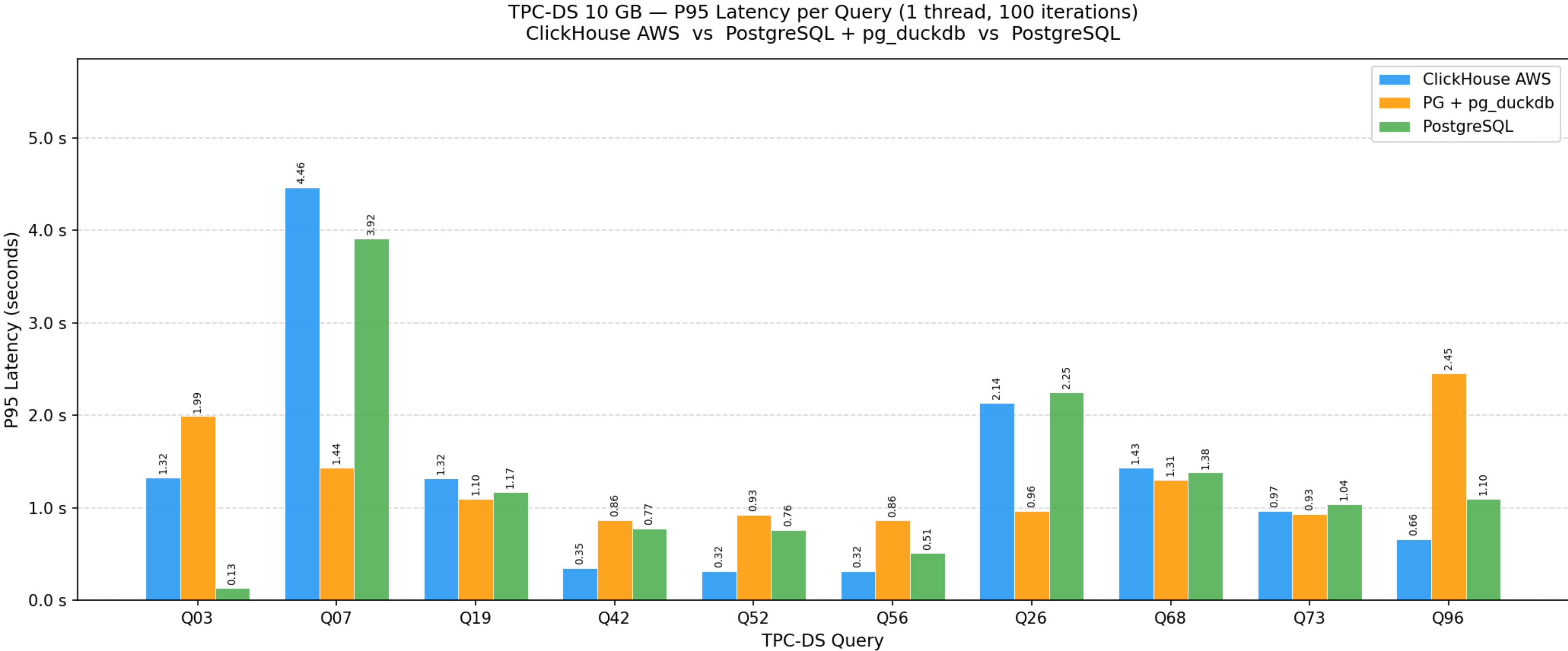
	Query	pg_duckdb	PG	Factor
✓ Best	Q14	19.3s	40.0s	2.07x faster
✓ Best	Q16	13.5s	22.7s	1.68x faster
✗ Worst	Q26	28.1s	0.003s	~9,150x slower

	Query	pg_duckdb	PG	Factor
✓ Best	Q12	177s	611s	3.45x faster
✓ Best	Q34	522s	793s	1.52x faster
✗ Worst	Q25	522s	0.001s	~560,000x slower
✗ Worst	Q27	521s	0.002s	~311,000x slower

TPC-DS Data Set Setup

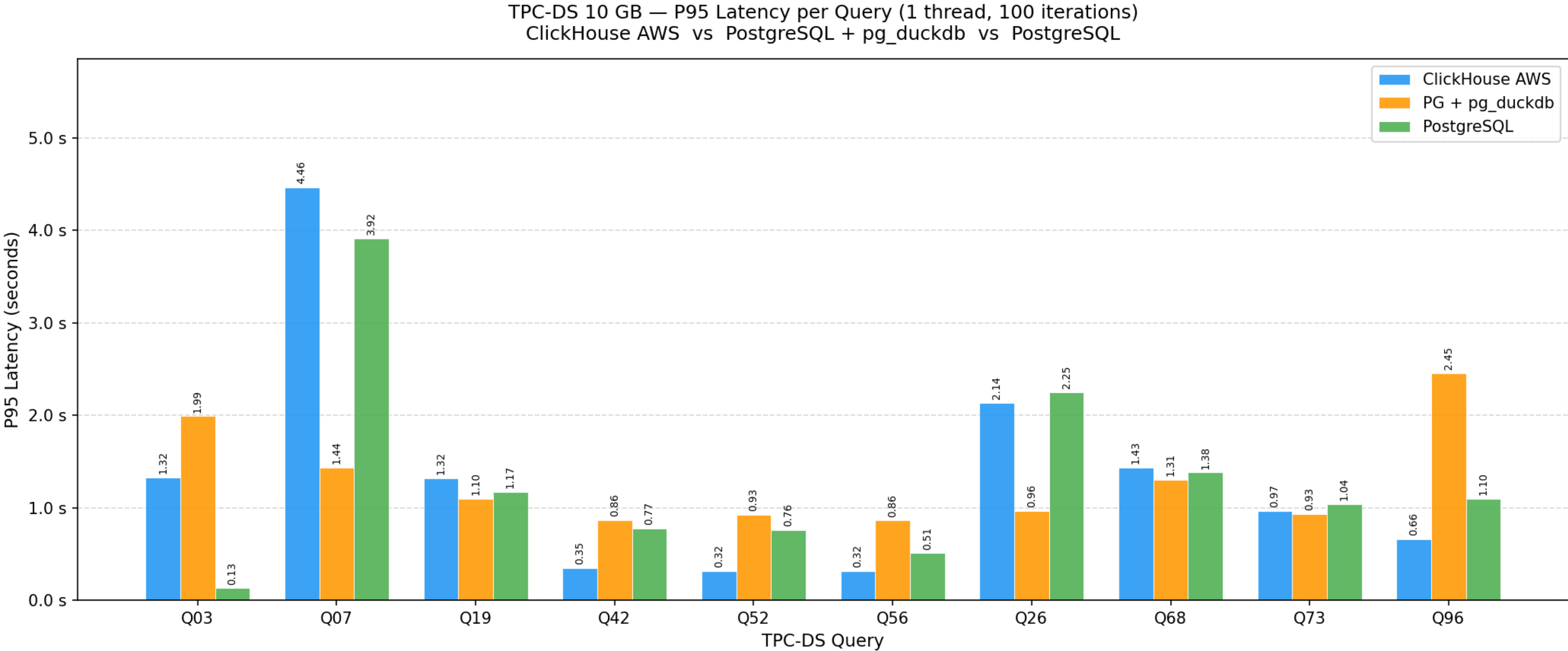
- Analytics domain: data warehouse
- Data set size is 10GB (scale factor 10)
- 24 tables in snowflake schema
- Picked 10 out of 99 official TPC-DS queries with multi-way joins, aggregations, and grouping
- Single threaded and sequential query execution
- 1000 iterations per query

TPC-DS — Benchmark Results



	Clickhouse	pg_duckdb	PostgreSQL
✔ Wins	4	2	1
Draws	3		

TPC-DS — Benchmark Results



pg_duckdb Perspective

	Query	ClickHouse	pg_duckdb	PostgreS	Factor
✓ Best	Q07	4.46s	1.44s	3.92s	2.73x faster than closest
✗ Worst	Q96	0.66s	2.45s	1.10s	2.23x slower than fastest

What We Skipped (for Now)

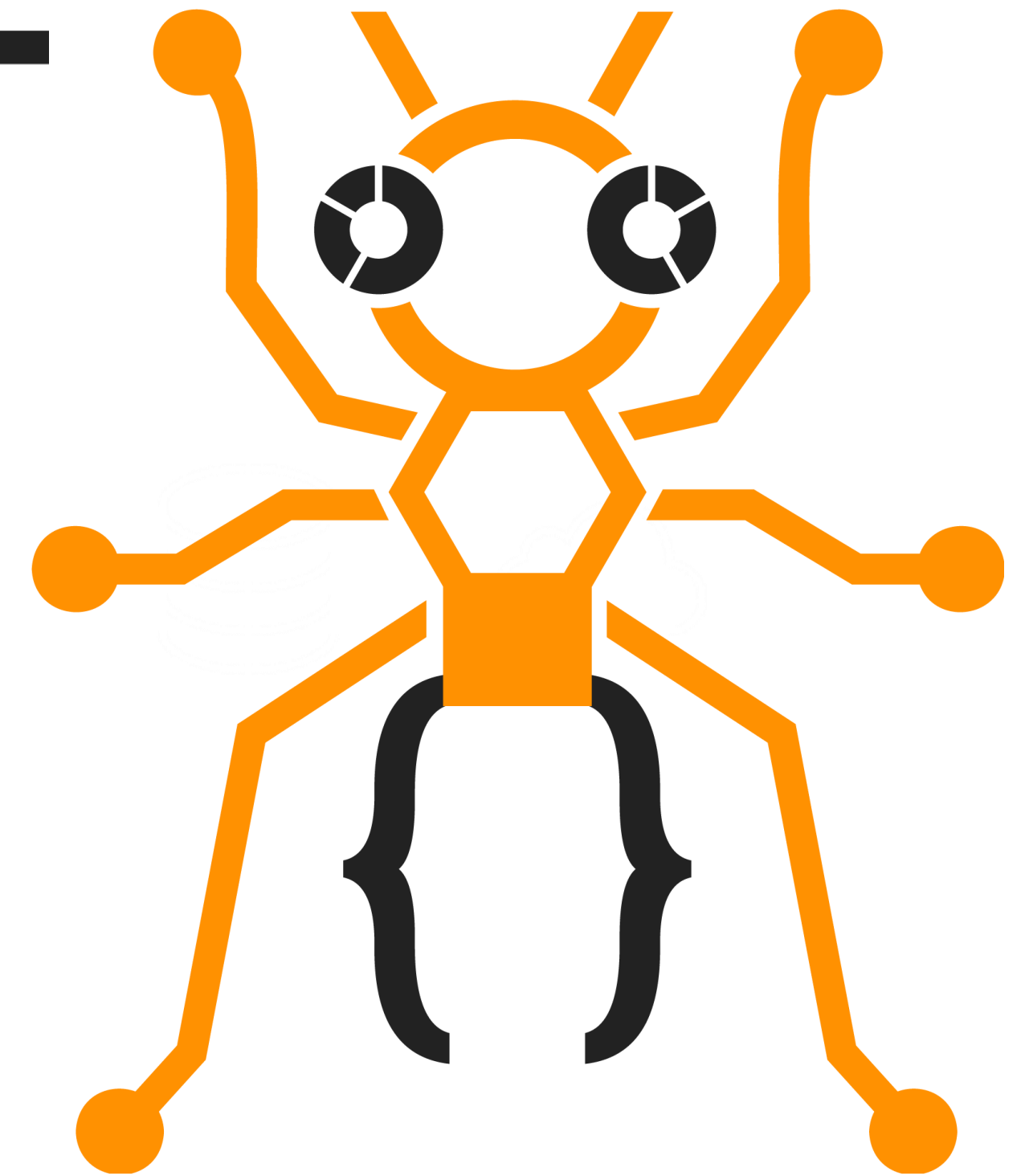
- Looking into the execution plans
- Running an optimal benchmark for PostgreSQL by applying pg_duckdb only for the relevant queries
- Larger TPC-DS data sets
- Concurrent query execution
- Make use of the pg_duckdb data lake and cloud analytics features
- Benchmarking other columnar extension for PostgreSQL

Is PostgreSQL Eating Analytics Too?

- PostgreSQL is expanding beyond OLTP
 - A growing ecosystem of columnar extensions
 - Major analytics vendors increasingly position PostgreSQL as the operational layer
- There is no single "analytical PostgreSQL extension"
 - Citus, TimescaleDB and pg_duckdb and many more target different workload classes
- pg_duckdb shows promising but mixed results
 - Significant performance gains for selected queries
 - Large regressions for others indicate that workload characteristics matter
- Benchmark road ahead for PostgreSQL Analytics
 - More performance studies on additional columnar extensions
 - Analyze more analytical workload shapes

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Thank you!



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